Maximum Sum Submatrix

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The problem

- Largest sum among all size x size square submatrices, using 2D prefix sums.

Intuition

- Two-dimensional prefix sums give any rectangle's sum in constant time.

Approach

- 1. Build prefix sums
- 2. For each bottom-right corner, compute the size by size square sum
- 3. Track the maximum

Java solution

```java
public class MaximumSumSubmatrix {
    public static int maximumSumSubmatrix(int[][] m, int size) {
        int R = m.length, C = m[0].length;
        int[][] p = new int[R + 1][C + 1];
        for (int r = 0; r < R; r++) for (int c = 0; c < C; c++) p[r + 1][c + 1] = m[r][c] + p[r][c + 1] + p[r + 1][c] - p[r][c];
        int best = Integer.MIN_VALUE;
        for (int r = size; r <= R; r++) for (int c = size; c <= C; c++)
            best = Math.max(best, p[r][c] - p[r - size][c] - p[r][c - size] + p[r - size][c - size]);
        return best;
    }
}
```

Complexity

- Time: O(w*h)
- Space: O(w*h)

Recap

- Two-dimensional prefix sums give any rectangle's sum in constant time.
- Complexity: O(w*h)